Deploying Static Web Projects via GitHub & Netlify

A Complete Developer Guide to Repository Setup, Configuration, and Automated Continuous Deployment

Project Documentation & Architecture Overview

1. Project Structure & Files

Understanding the standard static website layout configured for GitHub hosting

Frontend Source Files

- `index.html` — The standard entry document containing semantic HTML5 markup.
- `assets/style.css` — Centralized stylesheet containing global design tokens and responsive styles.
- `assets/app.js` — JavaScript ES6 application module managing dynamic DOM interactions.

Deployment & Metadata

- `netlify.toml` — Global configuration file instructing Netlify build output and rewrite rules.
- `_redirects` — Simplified URL fallback directives for single-page routing support.
- `README.md` — Comprehensive developer instructions for cloning, setup, and deployment.

2. Netlify Redirect & Build Rules

Configuring proper rewrite directives to handle Client-Side Routing and SPA fallbacks

`netlify.toml` Configuration

Specifies the root directory publish path and redirect configuration:

```
[build]
  publish = "."

[[redirects]]
  from = "/*"
  to = "/index.html"
  status = 200
```

`_redirects` Rule Syntax

Provides a direct fallback mechanism to ensure sub-paths return 200 HTTP status:

```
/*    /index.html   200
```

Using both guarantees compatibility across different Netlify engine deployment targets.

3. GitHub Repository Initialization

Step-by-step terminal commands to commit local project files to remote origin

CLI Initialization Steps

```
# 1. Initialize Git in the extracted project folder
git init

# 2. Stage all assets, configuration files, and documentation
git add .

# 3. Create initial commit snapshot
git commit -m "Initial commit: Add application code and deployment config"

# 4. Set main default branch and link remote repository
git branch -M main
git remote add origin https://github.com/your-username/your-repo-name.git

# 5. Push code to GitHub repository
git push -u origin main
```

4. Linking GitHub to Netlify (CI/CD)

Automated continuous delivery pipeline configuration setup

Step 1: Connect Account

Log into Netlify dashboard, select **Add new site** → **Import an existing project**, and authorize GitHub connection.

Step 2: Set Build Config

Select your repository. Set **Publish directory** to `.` and leave the **Build command** field blank.

Step 3: Auto Deployments

Click **Deploy Site**. Netlify automatically builds and deploys on every `git push` to main.

5. Summary & Best Practices

Key takeaways for managing project updates and production deployments

Category	Recommendation	Impact
Version Control	Keep `README.md` updated with changes	Improves documentation for collaborators
Redirects	Keep `_redirects` and `netlify.toml` in root	Prevents 404 page errors on page refresh
Assets	Use relative path references (`assets/app.js`)	Ensures styles load correctly on custom subdomains
Continuous Integration	Push changes directly to main branch on GitHub	Triggers instant automatic deployment on Netlify